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RETROFIT REPORT · CONDITION, MOISTURE, VENTILATION AND INSULATION, WITH A PLAN FOR THE WORKS

Anonymised sample report

Why the house is cold and mouldy, what it needs, and how to fold the fixes into the extension, new kitchen, underfloor heating and a possible heat pump. Built from the room-by-room inspection, humidity and damp readings and extract fan tests carried out on 21 October 2026.



PREPARED FOR	VISIT	ISSUED	ASSESSOR
The client	Wednesday 21 October 2026, 9.00am	22 October 2026	Dominic Bowkett

01 At a glance

Thank you both for showing me round on Wednesday. You told me what you want from the house: a kitchen that is warm enough to use all year, and an end to the cold, the condensation and the mould. This report explains why the house behaves as it does, and how to put it right as part of the extension and kitchen work you are already planning with the architect. The architect shared his planning drawings before the visit, and section 11 shows how this report fits them.

As agreed, this is a retrofit report rather than a formal energy rating. You are about to change the shape of the house, so an energy model of the house as it stands today would be out of date within a year. Instead, the report concentrates on what I found on the day: the defects, the moisture and humidity readings, the ventilation, the insulation that is missing, and the order in which to tackle it all.

The short version: the house is losing heat downwards, upwards and through its windows, and it has almost no working ventilation to carry moisture away. The two problems feed each other. Cold surfaces plus damp air give you condensation, and condensation gives you mould. The good news is that almost everything I found is fixable, and much of it fits naturally into the work you are already planning.

What I found

01 **The air in the house is very damp.**
Relative humidity was **91.6% on the landing** (at 12.0°C) and **81.8% in the hall** (at 14.1°C). A healthy home sits at 40–60%. At those levels any surface colder than about 11°C, such as a window frame, a corner or a cold ceiling, will run with condensation. The heating had been off for two weeks and it was a wet day, so these are worse than usual, but they are consistent with the mould you see all year round.

02 **Your bedroom has damp at the window reveals and dormer cheeks, and mould on the ceiling.**
The damp meter read in the wet range (304 and 354) on the walls beside the dormer windows, where the walls are coldest, and borderline (up to 190) on the ceiling near the windows. There is black mould spotting on the ceiling and around the window frames. I believe this is mainly condensation rather than a leak, but the dormer roofs and flashings should be checked after heavy rain to rule out water coming in from outside.

03 **The back corner of the kitchen is being wetted from both sides.**
Outside, the decking sits tight against the wall, a downpipe discharges at the corner, rain splashes back onto the render, mortar is missing in the stonework and the ventilation grille serving that corner has been permanently sealed on the inside. Inside, the reveal beside the back door read wet at low level, the paint is flaking and the dresser and cupboards against that wall have mould behind them.

04 **There is no effective extract ventilation anywhere in the house.**
I tested the three bathroom fans: **6.6 l/s** in your en-suite (which has no window), **2.5 l/s** in the family bathroom and **9.2 l/s** in the Bedroom 2 en-suite, against a minimum of 15 l/s. The kitchen has a recirculating hood that puts the steam straight back into the room, and the utility has no extract at all. Very few windows have trickle vents and none of the rooms meets today's minimum for background ventilation. The gaps under the doors are too small on almost every door, so even a good fan could not draw air through the house.

05 **The floor over the cellar and void is the biggest single heat loss.**

The ground floor is suspended timber over a walk-in cellar and a large crawl void. Under the kitchen there is no insulation at all. Under the rest, the old mineral wool between the joists has largely fallen out onto the ground. The floor is effectively open to outside air, which is why the kitchen feels cold from the feet up. You also have the perfect opportunity to fix it: the whole floor can be insulated from below without lifting a single board.

06 **The loft has too little insulation, and it is patchy.**

The insulation over the flat ceilings is uneven: about 150–200 mm where I measured, much thinner or missing elsewhere, around 100 mm overall, against the 270 mm recommended today. The loft hatch is not insulated, and the fan ducts that run across the loft are not insulated either. What is in the sloping ceilings and dormer cheeks of the rooms in the roof could not be seen.

07 **The windows are old and many of them are running with condensation.**

They are uPVC double glazing with a 16 mm gap. Apart from those in the 2010 extension, they are of unknown age, with no visible low-emissivity coating or warm-edge spacer bars. There is condensation and mould on the frames in most rooms. The front entrance has single-glazed top lights over the doors. You are already replacing about half of the windows; my advice is to do them all.

08 **The office extension and the conservatory are essentially unheated, uninsulated spaces.**

The office walls are timber frame about 200 mm thick, with little or no insulation, and there is no fixed heating. The conservatory has a glazed roof and only a portable heater. Both are coming down in the architect's scheme, which is the right call.

09 **The cracks in your bedroom should be looked at by a structural engineer before the works.**

There is a crack running down from the ceiling near the bedroom door, and another above the doorway. You told me they have grown over five years. They are where the 2010 extension meets the original house, so they may be settlement, but it is worth knowing for certain before you redecorate and rebuild.

The five things I would do first

#	ACTION	INDICATIVE COST
1	**Fix the ventilation now, not after the building work.** Replace the three bathroom fans with quiet, continuously running extract fans (dMEV), insulate their ducts in the loft, and trim every internal door to leave a 10 mm gap.	£900–£1,600 fans £20–£40 a door
2	**Stop the water at the back kitchen corner.** Take the downpipe into a drain or away from the wall, pull the decking back from the wall and repoint the missing mortar. The old grille is permanently sealed, so the kitchen's ventilation needs to come from elsewhere (section 06).	£300–£900
3	**Get the heating back on and keep it steady.** Once the boiler is repaired, run the house at a low, even temperature rather than switching rooms off, and dry laundry with a dehumidifier rather than an open window in Bedroom 3.	£150–£250 dehumidifier
4	**Insulate the whole floor from below**, cellar and void, in one go as part of the works, with rigid foam board between the joists and an airtight finish. It is the biggest improvement to the kitchen's comfort, and essential under underfloor heating.	£6,000–£12,000
5	**Have a structural engineer look at the bedroom cracks** before the design is finalised, and top up the loft to 270 mm with an insulated hatch.	£350–£600 engineer £800–£1,800 loft

Costs throughout this report are indicative 2026 ranges for the South East, to help with budgeting. They are not quotes.

02 Your questions, answered

You raised a number of questions on the day. Here are straight answers to each.

“Why is the house so mouldy, and why is it worst in our bedroom and the kitchen?”

Mould needs three things: moisture in the air, a cold surface for it to condense on, and time. The house has all three. Two people, cooking, showers and laundry put several litres of water into the air every day. With the extract fans moving only a fraction of the air they should, closed windows and doors with no gap underneath, that moisture has nowhere to go, so the humidity climbs to 80–90%. It then condenses on the coldest surfaces: the window frames, the dormer cheeks and ceiling of your bedroom, and the back wall of the kitchen.

Your bedroom is in the roof, with sloping ceilings and dormers that are probably thinly insulated, and it shares a wall with an en-suite that has no window and a weak fan. The kitchen is the biggest room over the coldest floor, its back wall is being wetted from outside, and the cooker hood recirculates the steam. That is why those two rooms are the worst.

“Will underfloor heating work, given the void underneath?”

Yes, and the void is an advantage, not a problem. Underfloor heating gives out heat in both directions, so on an uninsulated suspended floor a large share of it would go straight down into the cellar. With the floor properly insulated from below, the heat goes up into the room, where you want it. The void gives the installer full access to fit the insulation and run the pipework without disturbing the rooms above. Underfloor heating also runs at low water temperatures, which makes it an excellent partner for a heat pump.

“Should we insulate from below while we have the space?”

Yes. It is the single most valuable thing you can do for the kitchen. Rigid foam boards (PIR, such as Celotex) cut tightly between the joists and sealed at every edge will do the job. PIR is roughly twice as effective as the same thickness of mineral wool, so 100 mm of board does the work of about 200 mm of quilt. I would aim for 100–150 mm, more under the underfloor heating zones, with an airtight layer so that cold air cannot wash round the boards. The void must stay ventilated: never block the air bricks to “keep the cold out”. Details are in section 07.

“Could an air source heat pump work in a house like this?”

Yes, once the fabric has been improved. Construction-wise the house behaves like a modern one rather than a traditional solid-walled cottage, and I visit plenty of houses this size running happily on heat pumps. The key is to insulate the floor, top up the loft, replace the windows and sort the ventilation first, so the heat pump can be sized for the finished house rather than the leaky one. A heat pump installer, or a heat loss specialist I can put you in touch with, will need to do a room-by-room heat loss calculation based on the post-works specification. That will confirm the size of the unit, whether any upstairs radiators need upsizing, and the cylinder. More in section 10.

“We are replacing about half the windows. Should we do them all?”

My advice is to do them all, ideally at the same time as the building work; only the newer windows in the 2010 extension might be worth keeping. The older units have no low-emissivity coating or warm-edge spacers, and the frames have aged. Mould and condensation on them is widespread. Doing the whole house at once gives you one installer, one standard and one set of trickle vents designed together with the ventilation, and you avoid a second round of disruption and making good. The single-glazed top lights over the front doors should be replaced when the hall is refurbished.

One important point: new windows make a house more airtight. That is good for heat, but it removes some of the accidental ventilation the house relies on today. That is why the ventilation upgrades have to come first, or at the latest at the same time.

“Do we need mechanical ventilation?”

You need effective extract ventilation in the wet rooms and kitchen at the very least, and that is mechanical by definition. The simplest and least disruptive approach is a quiet, continuously running extract fan in each wet room and the kitchen (known as dMEV), boosting automatically when humidity rises. These cost a few pounds a year to run and are quiet enough to leave on. Given how damp the house is, I would also consider a positive input ventilation unit (PIV) in the loft, which gently pushes filtered air into the landing and helps dry the upstairs. A full ducted system with heat recovery (MVHR) is the gold standard but is disruptive to retrofit, and only really pays off if the whole house is made very airtight. Section 06 compares the options; it is a decision to make with the architect.

“Could we put insulation on the inside of the walls, or clad the outside?”

Both are possible, but neither is my first priority. The main walls measure about 280–300 mm and are most likely cavity walls. The first step is to find out what is in the cavity with a borescope; on an exposed hillside like this, wind-driven rain needs care before any cavity fill. Internal insulation suits the dry rooms but carries real condensation risk in the kitchen and bathrooms. External insulation would change the appearance of the house, and because the house is in the High Weald National Landscape it would need planning permission: cladding the outside of a house there is not permitted development. Floor, loft, windows and ventilation will give you far more for the money first.

“Is the crack in our bedroom something to worry about?”

Probably not in itself, but I would not ignore it. The main crack runs diagonally from the ceiling down the wall near your bedroom door, with a second one above the doorway, at the junction between the 2010 extension and the original house. Settlement cracks at junctions like this are common, but you told me it has been getting wider for five years, and in places it is approaching 5 mm. With major works about to start, have a structural engineer look at it now so that any cause can be dealt with in the architect's design. Section 09 lists some local practices.

“Will the new extension solve the cold and the leaks?”

It will solve some of them. The architect's scheme replaces the office and the conservatory, two of the worst parts of the building, with new construction built to today's Building Regulations, replaces every window, puts the damp kitchen corner inside the new extension and adds a heat pump. It does not yet cover the floor under the rest of the house, the loft, the rooms in the roof or the ventilation. Section 11 lists what to add to the specification, and the plan in section 13 sets out the order.

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03 The house and how you use it

Type	Detached chalet-style house on a sloping site, originally built in the early 1920s, with rooms in the roof lit by dormers. Two-storey extension at the west end, about 2010 (living room with the principal bedroom over). Dormers added in stages by the previous owner. Flat-roofed, timber-framed office extension at the front (date unknown) and a uPVC conservatory with a glazed roof off the dining room at the rear.
Floor area	About 255 m² across the two floors (magicplan survey, internal), including the office extension and conservatory. Ground floor: living room, dining room, snug, hall, WC, kitchen/breakfast room and utility. First floor: principal bedroom with en-suite, three further bedrooms (Bedroom 2 with en-suite, Bedroom 4 used as an office), family bathroom and landing.
Walls	Main house rendered, about 280–300 mm measured at the window reveals, most likely cavity walls (fill unknown). 2010 extension assumed built and insulated to 2010 Building Regulations. Office extension timber frame, about 200 mm, little or no insulation. Rooms in the roof timber-framed, with tile-hung dormer cheeks. Exposed stone plinth to the lower ground floor at the rear.
Floors	Suspended timber throughout, over a walk-in cellar (east end) and a large crawl void that tapers with the slope. No insulation under the kitchen; mineral wool between the joists elsewhere, much of it fallen.
Roof and loft	Pitched clay tile roof with tile-hung dormers, heavy moss. Loft over the flat ceilings with patchy mineral wool, about 100 mm overall. Uninsulated loft hatch and fan ducts. Sloping ceilings, dormer cheeks and boarded eaves could not be inspected.
Windows and doors	uPVC double glazing, 16 mm gap. Most are of unknown date with no visible low-E coating or warm-edge spacers; those in the 2010 extension are newer, with the darker spacer bar. French doors to the living room (front and rear), dining room and snug. Front entrance doors with single-glazed top lights, cat flap and letterbox.
Heating and hot water	Boulter Camray Utility floor-standing oil boiler in the cellar, an older model and most likely non-condensing (leaking at the time of the visit, repair due Monday 26 October), with radiators and thermostatic valves and separate heating and hot water programmers. Unvented (mains-pressure) hot water cylinder in the cellar. Oil tanks at the side. Closed room heater (log burner) in the living room. No fixed heating in the office extension or conservatory. Portable heater used in the upstairs office. Bottled gas for the range cooker.
Ventilation	Extract fans in the three upstairs wet rooms, all well below the required rate. Recirculating cooker hood in the kitchen. No extract in the utility or WC. Few trickle vents. Door undercuts too small on almost every door.
Electricity	Single-rate tariff, meter in the cellar. You have just bought an electric car.
Setting	Hillside site among mature trees. The rear (north) faces down the slope; trees shade the house and drop leaves and moss into the gutters. Flood zone 1. In the High Weald National Landscape; not in a conservation area.

PHOTO WITHHELD FOR PRIVACY

Front (south) elevation. Rooms in the roof with dormers, the 2010 extension on the left and the flat-roofed office extension on the right of the entrance.

PHOTO WITHHELD FOR PRIVACY

Rear (north) elevation. The conservatory off the dining room and the raised decking, with the cellar and void below the ground floor.



Kitchen/breakfast room. The largest room, over the coldest part of the floor.



Living room (2010 extension) with the log burner. Usually the warmest room.



Dining room, with French doors to the conservatory.



Hall, looking to the front entrance doors with their single-glazed top lights.



Principal bedroom, in the roof of the 2010 extension.

PHOTO WITHHELD FOR PRIVACY

Office extension from outside: flat roof, glazed on three sides.

How you use the house

The two of you live here with your dog. You moved in in January 2021 and are both now retired, so the house is occupied for most of the day. One of the owners works in the upstairs office (Bedroom 4), which gets cold and draughty and needs a plug-in heater. Most radiators are kept off most of the time; the living room, with the log burner, is warm and can overheat when it is full of people. The kitchen is the room you want to live in but cannot: it is so cold in winter that it is hardly used. One of you has a health condition that the cold makes noticeably worse, so comfort in the kitchen matters more here than in most houses.

Windows are rarely opened, except the small family bathroom window. The family bathroom is little used; you mainly use the en-suite. Laundry is dried indoors in Bedroom 3 with the window open. The cooker hood is not used. You are planning a major renovation with your architect, the architect. The scheme includes a new kitchen, living and dining extension at the rear in place of the conservatory, a replacement study for the office extension, a utility room in part of the snug, a new kitchen with an island and vented hob, underfloor heating, new windows throughout and an air source heat pump (section 11). The planning application has been made, and you are using the winter to get costs together.

04 Floor plan and where the problems are

The plans are drawn from the magicplan survey made on the day. The numbered markers show where the main findings are; each is described in the sections that follow. Windows are shown in blue, French doors dashed, and F marks an extract fan. The office and conservatory are shaded because they are outside the main heated envelope today. North is at the top; the plans are for orientation, so do not scale dimensions from them.



FLOOR PLAN WITHHELD FOR PRIVACY

1. **Kitchen back door reveal.** Wet at low level, flaking paint, mould behind the dresser and cupboards.
2. **Rear kitchen corner, outside.** Decking against the wall, downpipe, splashback, missing mortar, grille permanently sealed.
3. **Hall.** 81.8% relative humidity at 14.1°C. Single-glazed top lights over the front doors.
4. **Office extension.** About 200 mm timber frame, little or no insulation, no fixed heating.
5. **Conservatory.** Glazed roof, no fixed heating; to be demolished.
6. **Cellar and void** under the ground floor. No insulation under the kitchen; fallen insulation elsewhere.
7. **Dining room door.** Warped and sticking; undercut about 8 mm.
8. **Landing.** 91.6% relative humidity at 12.0°C. Loft hatch uninsulated.
9. **Principal bedroom.** Wet readings at the dormer reveals and cheeks; mould on the ceiling and window frames.
10. **Cracks** near and above the bedroom door, at the junction of the 2010 extension.
11. **Principal en-suite.** No window, fan 6.6 l/s, mould around the shower tray.
12. **Family bathroom.** Fan 2.5 l/s, mould on the walls and ceiling.
13. **Bedroom 2 en-suite.** Fan 9.2 l/s, the best of the three but still below 15 l/s.
14. **Bedroom 3.** Used for drying laundry with the window open.

05 Moisture, condensation and mould

This is the issue you asked me to look at most closely, and it is the one that drives most of the recommendations in this report.

LANDING HUMIDITY	HALL HUMIDITY	HEALTHY RANGE	CONDENSATION POINT
91.6%	**81.8%**	**40–60%**	**~11°C**
at 12.0°C, first floor	at 14.1°C, ground floor	relative humidity	surfaces colder than this get wet

Humidity

I measured the relative humidity with a Protimeter MMS3 hygrometer, left to settle in each location. It was **91.6% on the first-floor landing** at 12.0°C and **81.8% in the hall** at 14.1°C, with readings in the low 80s through the rest of the ground floor and 80–90% upstairs. Anything above about 60% for long periods allows mould to grow and dust mites to thrive. At 91.6% and 12°C, the air reaches its dew point at about 10.7°C, so any surface colder than that, such as a window frame, a cold corner or a poorly insulated ceiling, will become wet.

Two things on the day made the readings worse than usual: the heating had been off for about two weeks because of the boiler leak, so the house was cold, and it was a wet, still day outside. Even allowing for that, readings this high, together with the mould on the ceilings and frames, the clumping salt and sugar, and the peeling wallpaper you have seen, point to a house that holds far too much moisture for much of the year.

Damp readings

I used the same meter in non-invasive search mode, which reads moisture just below the surface. The readings are relative: below about 170 is normal for plaster, 170–200 is borderline, and above 200 indicates a wet surface.

LOCATION	READING	WHAT IT MEANS
Principal bedroom, wall beside the dormer windows (reveals and dormer cheeks)	**304, 354** and a further wet reading	WET Higher than I would expect. These are the coldest walls in the room, which suggests condensation on a poorly insulated surface. Check from outside after heavy rain to rule out water getting in at the dormer.
Principal bedroom, ceiling near the windows	160, 174, 181, 190	BORDERLINE Damp ceiling where the mould spotting is; condensation forming on the cold ceiling.
Principal bedroom, window head and board	137–172	Dry to borderline.
Principal bedroom, centre of ceiling	140	Dry.
Kitchen, reveal beside the back door	Wet range at low level; dry higher up	WET Water from outside at the base of the wall (decking, downpipe, sealed grille) combined with condensation. Paint and plaster flaking.



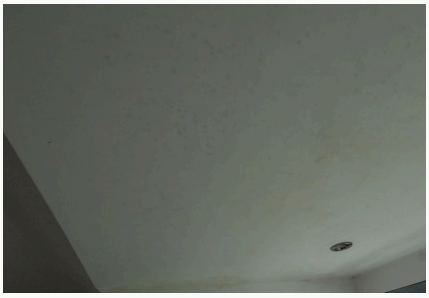
Landing. 91.6% relative humidity at 12.0°C.
10:41



Hall. 81.8% relative humidity at 14.1°C.
11:01



Principal bedroom dormer reveal. 354, in the wet range.
10:45



Principal bedroom ceiling. Black mould spotting across the ceiling over the window, coming back after cleaning.
10:43



Principal bedroom window. Mould and condensation staining on the frame and reveal.
10:43



Principal bedroom dormer reveal. 304, in the wet range.
10:45



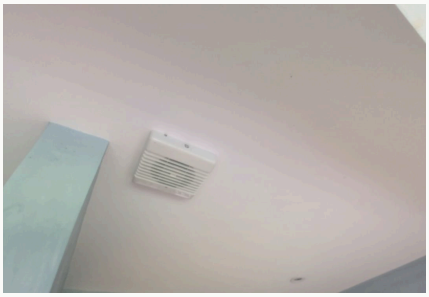
Kitchen back door reveal. Flaking paint and staining at low level.
11:04



Kitchen back door reveal. Reading in the wet (red) range.
11:04



Back door corner. The dresser and cupboards sit against the damp wall, where the mould is worst.
11:03



Family bathroom. Streaky mould staining on the walls and ceiling around the fan.
10:51



Principal en-suite. No window; the shower relies entirely on a weak fan.
10:47



Bedroom 3. Used for drying laundry, a large daily source of moisture.
10:50

Why it is happening

- **Too much moisture, too little ventilation.** Cooking, showering, drying laundry and breathing add moisture every day. The fans extract a fraction of what they should, the cooker hood recirculates, few windows have trickle vents and the doors are too tight to the floor for air to move.
- **Cold surfaces.** The coldest surfaces are the window frames, the dormer cheeks and sloping ceilings in the roof, and the walls over the uninsulated floor. That is exactly where the mould is.

- **Low, intermittent heating.** With most radiators kept off, rooms and surfaces stay cold, which makes condensation worse. A house heated steadily at a lower temperature is drier than one heated in bursts.
- **Water from outside at the back kitchen corner** (section 09), which adds to the condensation inside.

What to do now

- Clean the mould with a fungicidal wash (not just bleach, which bleaches the stain but leaves the roots) and repaint the bathroom, en-suite and bedroom ceilings with a mould-resistant paint. If it comes back quickly once the fans are working, that tells us there is more to find.
- Upgrade the fans and trim the doors (section 06). This is the most important single step.
- Once the boiler is fixed, keep the house at a steady, moderate temperature, around 18°C in bedrooms and 20°C in living spaces, rather than leaving rooms unheated.
- Dry laundry outside when you can. Indoors, use a dehumidifier in a closed room, or a heat-pump tumble dryer, rather than an open window in a cold room.
- Pull furniture a few centimetres off the outside walls, especially the kitchen dresser and the bedroom furniture, so air can reach the wall behind.
- A portable dehumidifier in the kitchen and your bedroom will help this winter while the bigger work is planned. If the washing machine stays in the current utility until the works, run the dehumidifier there with the door closed on washing days. Whether anything more is worth doing to that room depends on whether you keep it.

06 Ventilation

A house needs a steady supply of fresh air and a way to remove moisture at its source. At present the cottage has very little of either. In an older, leaky house much of this happens by accident through gaps; as you replace windows and insulate, those gaps close, so the deliberate ventilation has to be in place first or the condensation will get worse, not better.

ROOM	BACKGROUND VENTILATION	EXTRACT	DOOR UNDERCUT
Kitchen	No trickle vents	Recirculating hood only FAIL Needs 30 l/s ducted hood or 60 l/s fan	Insufficient; doors do not close well
Utility	Small window	None FAIL (washing machine). The architect's drawings move the utility into part of the snug; see below.	About sufficient
Living room	Trickle vents, about 200 mm in total	Not required. Closed room heater (log burner)	About 6 mm, insufficient
Dining room	No trickle vents	Not required	About 8 mm, slightly insufficient; door warped
Hall	No trickle vents; single-glazed top lights	Not required	n/a
WC	Opening window only	No fan	Insufficient
Principal bedroom	Trickle vents, about 200 mm in total	Not required	Insufficient
Principal en-suite	No window	**6.6 l/s** FAIL	Insufficient

Bedroom 2	Two trickle vents, both behind the blinds, so not working effectively	Not required	Insufficient
Bedroom 2 en-suite	Trickle vents in the window	**9.2 l/s** FAIL	Insufficient
Bedroom 3	None seen (blinds)	Not required	Just short
Bedroom 4 (office)	No trickle vents	Not required	Just short
Family bathroom	No trickle vents; window usually ajar	**2.5 l/s** FAIL	Insufficient
Landing	No trickle vents	n/a	To en-suite: insufficient

Fan rates measured with a Testo 417 anemometer and flow hood, fans on full speed. The intermittent minimum for a bathroom or shower room is 15 l/s (Approved Document F). Undercuts should give about 10 mm above the floor finish, equal to 7,600 mm² on a standard 760 mm door, so that air can reach the extract fans with the doors closed. None of the rooms meets today's minimum equivalent area for background ventilation.



Principal en-suite fan. 1.07 m/s, 6.6 l/s.
10:48



Family bathroom fan. 0.46 m/s, 2.5 l/s.
10:52



Bedroom 2 en-suite fan. 1.46 m/s, 9.2 l/s.
10:54



Principal bedroom door. Undercut well below 10 mm.
10:46



Dining room door. About 8 mm, and the door is warped.
11:00



Kitchen doors. Undercut insufficient; the pair do not close well.
11:04

The options

SYSTEM	HOW IT WORKS	FIT FOR THIS HOUSE
Intermittent fans (as now, but proper ones)	Fans that run when the light is on or on a humidistat, at 15 l/s or more.	Minimum. Easily switched off or ignored.
Decentralised MEV (dMEV)	A quiet fan in each wet room and the kitchen runs all the time at low speed, boosting with humidity. Wired so it cannot be switched off.	**Recommended**, now in the bathrooms and in the new kitchen and utility.
Positive input ventilation (PIV)	A unit in the loft pushes gently filtered, slightly tempered air into the landing, displacing damp air out through vents and gaps.	Worth considering alongside dMEV, given the humidity upstairs.

Central MEV or MVHR	Ducts from every room to a central unit in the loft; MVHR also recovers heat from the outgoing air.	Best performance, but disruptive and costly to retrofit. Only worth it if the house is made very airtight.

- **Bathrooms now:** replace the three fans with dMEV units. Insulate the ducts where they cross the loft (they are bare at the moment, so warm moist air condenses inside them and can drip back), keep the runs short and end them in a proper roof or wall terminal. Ask the electrician to fit an isolator outside each room, for servicing only.
- **Kitchen (new):** your idea of a vented downdraft hob on the island, ducted to outside through the void, is a good one, as long as it discharges outside and not into the void. Add a continuous dMEV in the kitchen as well, because the hob extract only runs when you cook.
- **Utility:** The architect's drawings take out the current utility and put a new one in part of the snug. That room appears to have no outside wall or window, so it must have a mechanical extract ducted to outside (a dMEV, or a 30 l/s fan), a door undercut for replacement air, and a heat-pump or vented dryer. There is no point spending money on a fan for the current utility, which is coming out.
- **Doors:** trim every internal door to a 10 mm undercut, measured from the finished floor on the higher side. Any carpenter can do it; allow around £20–£40 a door.
- **Trickle vents:** every new window should have them, sized to the room. Keep the existing ones open, especially overnight, and keep blinds and curtains clear of them. In Bedroom 2 both vents sit behind the blinds, so they do very little at the moment.
- **The log burner:** check with your installer whether it needs a permanent air vent for combustion air. That depends on its output and age, and becomes more important as the house gets tighter. Keep the carbon monoxide alarm tested.

07 Insulation

The ground floor, cellar and void

The whole ground floor is suspended timber: boards on joists over a walk-in cellar at the east end and a crawl void that tapers with the slope under the rest of the house. Under the kitchen and cellar there is no insulation at all; you can see the underside of the floorboards. Under the rest of the house there is mineral wool between the joists, held up with strips, but much of it has fallen onto the ground and what is left is slumped. There is no hatch between the cellar and the void, just an open access. In effect the kitchen and much of the ground floor sit over outside air, which is why the floor feels so cold.



Cellar. Joists and floorboards with no insulation beneath the kitchen.
10:34



Opening from the cellar into the void. Insulation hanging loose. There is no hatch.
10:35



Crawl void. Mineral wool between the joists, much of it fallen to the ground.
10:35

What I recommend: insulate the whole floor from below in one go, ideally while the builders are on site for the extension.

- Remove the old fallen and slumped insulation.
- Fit rigid PIR foam board (such as Celotex), 100–150 mm, cut tightly between the joists and supported on battens. Foam every edge, and tape the joints. Then fit an airtight layer under the joists so that cold air from the void cannot get round or behind the boards. Under the underfloor heating zones, aim for the thicker end.
- Cover the full extent of the floor. Insulating only part of it leaves a cold strip where the warm air will run to, and where condensation will form.
- **Keep the void ventilated.** The air bricks and grilles must stay open, and those covered at the front should be cleared, so the joists stay dry.
- Once the floor is insulated, the cellar will be colder, so lag the heating and hot water pipes running through it.

If the floor under the 2010 living room is a different construction (I could not get into that part of the void), the architect's team should check it when the works start.

The loft

The loft over the flat ceilings has mineral wool between and over the joists, but it is uneven. It measured about 150–200 mm where I put the tape in, but it is much thinner, compressed or missing in places, and averages around 100 mm. Today's recommendation is 270 mm. The fan ducts lie across the top of the insulation, uninsulated, and the loft hatch has no insulation. On the underside of the roof, a section of the timber boarding has been cut out and patched with new timber and membrane, and there is old staining on the boards nearby.



Loft. Uneven mineral wool, with the bathroom fan ducts lying bare across it.
10:38



Depth check. About 150–200 mm at this point; thinner elsewhere.
10:39



Underside of the roof. A patched section of the roof boarding, with staining on the boards nearby. MONITOR
10:38

- Top up to 270–300 mm over the flat ceilings, laid in two layers at right angles. If you need storage, use raised loft legs rather than compressing the insulation.
- Keep a clear ventilation gap at the eaves.
- Insulate the loft hatch and fit a compression seal.
- Insulate the fan ducts, or replace them with insulated ducting when the new fans go in.
- Check the patched area of roof boarding after heavy rain. If it is dry, no action is needed.

This is a quick, low-cost job that does not depend on the extension, so it can be done this winter.

The rooms in the roof

The first floor is built into the roof, with sloping ceilings, dormers and storage in the eaves. It is impossible to know how much insulation is in the slopes, dormer cheeks and stud walls without opening them up, and the eaves storage is fully boarded and full of belongings. The damp readings at the dormer cheeks in your bedroom suggest at least some of these surfaces are poorly insulated. The 2010 extension should have been insulated to the standards of the time, but bits are sometimes missed. The practical approach is to insulate the slopes and dormer cheeks properly the next time the roof is re-covered, or when any of those rooms is redecorated (for example, insulated plasterboard over the sloping ceilings, detailed with a vapour control layer).

The walls

- **Main house:** about 280–300 mm, most likely cavity construction. Whether the cavity is filled is unknown. A borescope check is the first step. On an exposed, wet hillside site, any cavity fill must be suitable for wind-driven rain.
- **2010 extension:** assumed insulated to 2010 standards.
- **Office extension:** about 200 mm timber frame, with little or no insulation. It is being demolished, so no work is needed.
- **Internal or external wall insulation:** possible later, but lower priority. Internal insulation in the kitchen and bathrooms needs careful moisture design. External insulation would need planning permission in the National Landscape, and must be detailed at the plinth, eaves and windows.



Wall thickness measured at a window reveal.
11:09



Loft hatch on the landing. Not insulated.
10:37



Eaves storage. Fully boarded and full; the insulation behind could not be seen.
11:18

08 Windows and doors

The windows are uPVC double glazing with a 16 mm gap between the panes and leaded-style bars. Apart from the windows in the 2010 extension, which have the darker warm-edge spacer bar of newer units, I could find no date stamp, no visible low-emissivity coating and no warm-edge spacer bars (the dark spacer found in units made from about 2002), so they are recorded as of unknown age. Many showed condensation, and there was mould on the frames and seals in most rooms, especially upstairs. Some opening sashes are awkward to use, and the kitchen windows are small and hard to open.



Principal bedroom. Condensation staining and mould at the frame.
10:43



Landing window. Double glazed, 16 mm, date unknown.
10:41



Front entrance. Single-glazed top lights over the doors, plus the cat flap and letterbox.
11:01

- **Replace all the windows and external doors** rather than half, ideally as part of the building works. The newer windows in the 2010 extension could be kept if they are in good order and the installer can fit matching trickle vents. Specify A-rated units (U-value 1.4 W/m²K or better, or triple glazing on the north side if the budget allows), with trickle vents sized for each room and secure night-vent positions.
- Fit them with care: insulate and seal the frame to the wall (an airtight tape inside, weatherproof outside), so the new windows do not leave a cold, leaky edge.
- Replace the front entrance doors and their single-glazed top lights with the hall refurbishment, with an insulated letterbox and draught-sealed cat flap.
- Use TrustMark and FENSA (or CERTASS) registered installers, get more than one quote, and coordinate the window order with the architect so the new openings in the extension match the rest.

09 Roof, exterior and structure

I inspected the outside from the ground in heavy rain; the weather was too wet to fly the drone. As I am local, I am happy to come back on a dry, calm day and do the drone survey of the roof for you. Each item is marked ACT NOW (before winter or before other work), PLAN (as part of the works or at the next opportunity) or MONITOR.

PHOTO WITHHELD FOR PRIVACY

Rear kitchen corner. The decking is built tight against the wall, with a downpipe at the corner and missing and eroded mortar in the stone plinth. ACT NOW

10:35

PHOTO WITHHELD FOR PRIVACY

Same corner, closer. Flaking render, staining and open joints where water is being held against the wall. ACT NOW

10:35

PHOTO WITHHELD FOR PRIVACY

Ventilation grille above the decking. The grille serving this corner has been permanently sealed on the inside, so the ventilation has to be provided another way.

PLAN

10:35

PHOTO WITHHELD FOR PRIVACY

Roof. Heavy moss on the clay tiles; I watched water pouring off the roof by the dormers in the rain. No missing tiles seen from the ground. PLAN

10:33

PHOTO WITHHELD FOR PRIVACY

Dormers and gutters. Moss and leaves collect in the gutters, which overflow in heavy rain. Keep them clear, especially above the principal bedroom. ACT NOW

10:32



Vent grille and downpipe. Keep vents clear and the downpipe connected to a drain.

PLAN

10:33

PHOTO WITHHELD FOR PRIVACY

Base of the render, front. A crack at the render-to-brick junction, and the render taken down close to the paving. PLAN

10:32

PHOTO WITHHELD FOR PRIVACY

Under the kitchen bay window. A vertical hairline crack in the render. MONITOR

10:32

PHOTO WITHHELD FOR PRIVACY

Under the decking. The cellar door and the stone plinth stay damp and shaded here.

PLAN

10:33

The rear kitchen corner

This corner is a textbook damp trap. The decking is built against the wall, so rainwater and splashback sit on the render and soak in. A downpipe discharges right there. Mortar is missing from the stone below. The grille that once ventilated the corner has been permanently sealed on the inside, and there is nothing to be gained by trying to reopen it. Inside, this is exactly where the wet readings, the flaking paint and the worst of the kitchen mould are. Before any insulation goes into this part of the house, and before the new kitchen is fitted against it:

- Take the downpipe into the drain, or extend it well clear of the wall.
- Cut the decking back to leave a clear gap of at least 50 mm from the wall, with no boards bridging the render.
- Repoint the missing and eroded joints with a mortar that matches the stone (lime-based on the old stone plinth).

- Plan the kitchen's ventilation differently instead of relying on the old grille: a continuous dMEV unit, the ducted extract from the hob, or another form of mechanical ventilation (section 06).

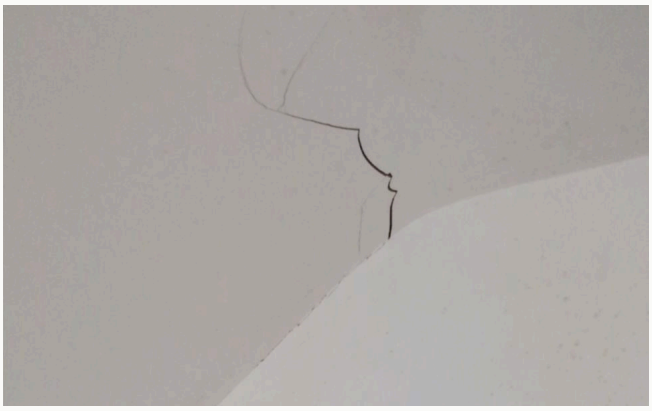
In the architect's scheme this corner ends up inside the new kitchen, living and dining extension, and the decking is replaced by a paved terrace. That removes the cause, but the wall and its stone base must be dried out and repaired before they are enclosed, and the terrace levels and drainage need detailing (section 11, item 8).

Cracks in the principal bedroom



Crack near the bedroom door, running down from the sloping ceiling. ENGINEER

10:43



Crack above the doorway, in the same area. ENGINEER

10:45

Both cracks are at the junction between the 2010 extension and the original house. Cracking where new meets old is common and often harmless settlement or plasterboard movement, and from what I saw I am not overly concerned. However, you told me the main crack has been getting worse for five years, and in places it is approaching 5 mm wide, which is the point at which I would always want a structural engineer's opinion. With major works planned, have it checked now so that any cause can be designed out. Some local practices are listed below.

PRACTICE	CONTACT	NOTES
SNGT Consulting East Grinstead	01342 420559 info@sngtconsulting.co.uk	Structural engineering consultancy, mainly house alterations, extensions and loft conversions, which would also suit the extension. Ask whether they will inspect and report on the cracks.
Romala Design Tunbridge Wells, also Battle and Bexhill	01892 240128 romaladesign.co.uk	Civil and structural engineers. Structural and subsidence inspections and reports on domestic property.
Abbott Holliday London and the South East; West Sussex branch at Billingshurst	0330 133 8407 enquiries@abbott-holliday.co.uk	Chartered structural engineers. Inspect existing buildings for structural movement, cracks and defects, and report on the cause and the remedy. Check they cover your postcode.
Owens Consulting Engineers South coast; covers East and West Sussex	01903 774844 enquiries@owens-consulting.co.uk	Chartered structural and civil engineers. Structural inspections and reports, including cracking in houses.

I have no connection with any of these practices. Ask for a fixed fee for a visual inspection and a short written report on the cracks, and check the engineer is chartered (CEng MIStructE). It is also worth asking the architect whether he already works with an engineer, because the extension will need one anyway.

Until then, you can monitor it simply: pencil a short line across each end and the date, and photograph it with a ruler every couple of months.

10 Heating, underfloor heating and heat pumps



Boulter Camray Utility oil boiler in the cellar, opened up for the leak repair at the time of the visit.
10:34



Unvented hot water cylinder in the cellar, with its expansion vessel and inlet controls on the mains supply.
10:34



Log burner (closed room heater) in the living room.
10:59

Now

The heating is a floor-standing oil boiler in the cellar with radiators, thermostatic radiator valves and separate programmers for heating and hot water. The boiler was leaking and switched off at the time of the visit, with parts due to be fitted on Monday 26 October. The oil tanks are at the side of the house. Once it is repaired:

- Run the house at a steady, moderate temperature, rather than leaving most of the radiators off. It is better for comfort, and for condensation.
- Fit smart radiator valves or a zoned controller, so that rooms such as the home office can be heated when they are in use without heating the whole house.
- Have the boiler serviced annually, and keep it running safely until the heat pump decision is made.

Underfloor heating

Underfloor heating in the new kitchen and dining room is a good idea, and including the living room would avoid regret later. Doing the ground floor in one go gives you a continuous floor and frees wall space. On a suspended timber floor it is fitted between or on top of the joists with aluminium spreader plates, or as a low-profile overlay. It must sit over the floor insulation described in section 07, otherwise much of the heat will go down into the cellar. Underfloor heating works at low water temperatures (around 35–45°C), which suits a heat pump well. The living room's log burner can stay as a secondary heat source.

An air source heat pump

A heat pump is suitable for this house once the floor, loft, windows and ventilation have been improved. Points to plan for:

- **Heat loss calculation first.** The installer, or an independent heat loss specialist, needs to carry out a room-by-room calculation to MCS standards, based on the specification after the works, not the house as it is today. It confirms the size of the heat pump, which upstairs radiators need upsizing, and the underfloor heating design. I can introduce you to people who do this.
- **Hot water.** Heat pumps need a cylinder with a large heating coil. Your existing unvented cylinder may not have one, so the installer will check whether it can stay; if not, the cellar is a good place for a new one. Unvented cylinders need an annual safety check, which can be done with the boiler service.
- **Outdoor unit.** The architect's drawings put it on the east wall beside the new extension, close to where the oil tanks are now, which is the natural place once the tanks are removed. It needs space around it for air flow and access, a drain for the defrost water, and a noise check at the nearest neighbour.
- **Costs and grants.** Typically £14,000–£20,000 for a house this size, before the Boiler Upgrade Scheme grant, through an MCS-certified installer. The grant is £7,500 as standard, but for homes heated by oil, like yours, it is currently £9,000, until 31 March 2027. Your works may not be finished by then, but it is worth knowing about if the timing works, and the scheme may change again. Like a boiler, a heat pump has a service life of around 20 years or more and needs annual servicing.

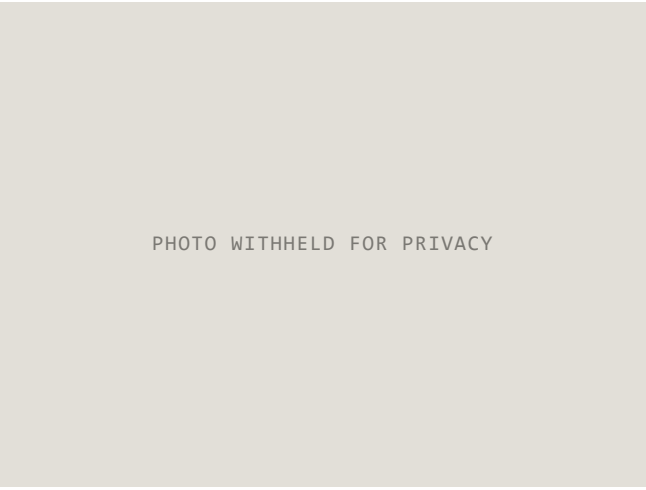
- **Electricity tariff.** With an electric car now and a heat pump later, a time-of-use or EV tariff (which needs a smart meter) will cut running costs considerably compared with your current single-rate tariff.
- **Staying on oil.** This remains an option in the short term, but oil prices are volatile, and a new oil boiler now would tie you in for 15 years or more. If the heat pump will follow the works within a year or two, I would repair rather than replace the oil boiler in the meantime.

11 The architect's proposals and how they fit

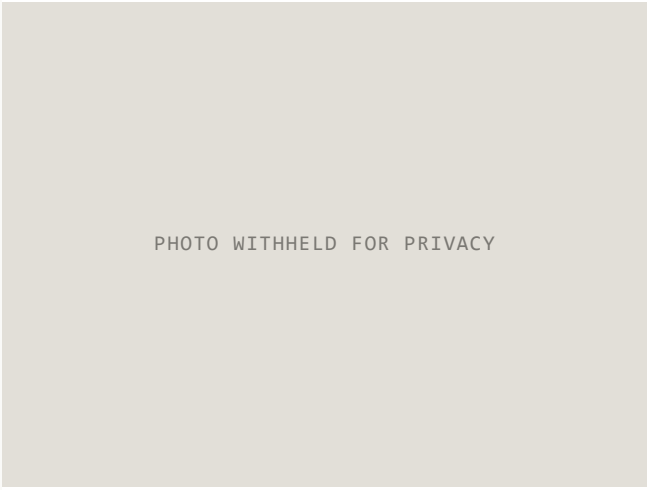
Before the visit, the architect shared the planning drawings for the works and the Design and Access Statement, and I took them into account during the inspection. The drawings are project 2613, revision P1, issued for planning on 27 August 2026, and the statement is dated 4 September 2026. This section summarises the scheme, sets out what it already does for the problems in this report, and lists what I would add to the specification so that the finished house is warm, dry and ready for a heat pump.

What the scheme proposes

- **A new rear extension behind the kitchen.** It replaces and relocates the massing of the conservatory to make an open-plan kitchen, living and dining room with an island. It has a red standing-seam metal roof with a fixed flat rooflight, aluminium sliding glazing to the north and a fixed picture window to the east. The walls are pale brick over a buff brick base, with access to the undercroft below.
- **A replacement front extension.** The poor-quality office is rebuilt as a study of about the same internal size, with thicker, insulated walls. It has a false-pitch tiled roof, white render over a stone base and eaves that line through with the main house.
- **Inside.** Part of the snug becomes the new utility room, with storage. The hall is opened up into an entrance hall, the dining room stays, and the living room's rear French doors become wider aluminium sliding doors. The ground-floor WC by the stairs is kept.
- **Outside.** The decking and the tall rendered wall at the rear are replaced by a paved terrace, stepped brick planters and gentler steps down to the garden, with metal railings.
- **Stair hall.** The rear stair gable is re-clad in clay tile hanging above the ground floor, and its eaves line is continued.
- **Energy.** An air source heat pump is shown on the east wall, "along with upgraded insulation internally". All the uPVC windows are replaced, and a new pitched rooflight serves the principal en-suite, which has no window today.
- **Unchanged.** The first floor and the main roof are generally unchanged. The internal floor area stays about the same overall.
- **Setting.** The site is in flood zone 1 and in the High Weald National Landscape.



Ground floor as existing. Lounge, dining room, snug, lobby, kitchen with its utility, the conservatory behind the dining room and the office at the front.
The architect 2613-DR-102 P1



Ground floor as proposed. The new kitchen, living and dining extension at the rear right, the utility in the former snug, the new study at the front, and the terrace and planters.
The architect 2613-DR-302 P1

North (garden) elevation as proposed. The new extension on the left, enlarged sliding doors, tile hanging on the stair gable, and the terrace and planters.
The architect 2613-DR-305 P1

East elevation as proposed. The heat pump at ground level, the picture window and the undercroft access below the new extension.
The architect 2613-DR-305 P1

First floor as proposed. Generally unchanged, with the new rooflight over the principal en-suite.
The architect 2613-DR-303 P1

South (front) elevation as proposed. The replacement study under its false-pitch roof, new windows throughout and the en-suite rooflight.
The architect 2613-DR-306 P1

Drawings © the architect, reproduced to coordinate this report with the design. Do not scale from them.

What the scheme already does for the problems in this report

- **The two worst spaces go.** The uninsulated office and the conservatory are replaced by new construction built to today's standards (sections 01 and 07).
- **Every window is replaced.** That matches my recommendation in section 08 to do them all rather than half.
- **A heat pump is part of the plan**, which suits the house once the fabric is improved (section 10).
- **The principal en-suite gets a rooflight.** That gives it daylight and a way to air it after a shower; it still needs a proper extract fan.
- **The decking is replaced by a paved terrace**, and the damp back corner of the kitchen ends up inside the new extension. That removes the decking and splashback that are wetting the wall today (section 09), provided the corner is repaired before it is enclosed.
- **The island layout** suits a vented downdraft hob ducted to outside, which you were already considering.

What I would add to the specification

None of these change the planning drawings. They belong in the Building Regulations drawings and the builder's specification, which is the natural next stage for the architect.

#	ADD TO THE SPECIFICATION	SEE
1	**A ventilation strategy.** It is not shown on the drawings, and it matters more once the new windows make the house tighter. • Continuous extract (dMEV) in the new kitchen, living and dining room, the ground-floor WC, the family bathroom and both en-suites. • The new utility in the former snug appears to have **no external wall or window**, so it needs a mechanical extract ducted to outside and a door undercut for replacement air. A heat-pump or vented dryer, never one that vents into the room. • The island hob extract ducted to outside, not into the undercroft. • Trickle vents, or equivalent frame vents, on every new window and sliding door, and 10 mm door undercuts throughout. • Consider positive input ventilation (PIV) for the upstairs.	Section 06
2	**Insulate the whole existing suspended floor from below**, not only the new extension's floor. The new kitchen, living and dining room sits partly over the existing kitchen floor and the void, and underfloor heating needs insulation beneath it everywhere it runs.	Section 07
3	**The new extension's floor over the undercroft.** Insulate it to Building Regulations standard (0.18 W/m²K or better) with an airtight layer. If the undercroft stays outside the heated envelope, make its access door weathertight and keep the void ventilated.	Section 07
4	**Pin down what "upgraded insulation internally" means.** I would put the loft top-up to 270–300 mm, the hatch, the floor and the windows first. If internal wall insulation is included, keep it to the dry rooms, use a vapour-control or breathable system as appropriate, detail the reveals and floor junctions, and avoid the kitchen and bathrooms unless a moisture risk assessment (BS 5250) supports it.	Sections 07, 14
5	**Insulate behind the new tile hanging on the stair gable.** That wall is being re-clad and scaffolded anyway, so adding insulation on battens behind the tiles (for example 60–100 mm of wood-fibre board) costs relatively little for the area. Detail the window reveals and the eaves so the cladding still fits.	Section 07
6	**New elements to current Part L.** Walls 0.18, roofs 0.15 and floors 0.18 W/m²K; windows and doors 1.4 W/m²K, with thermally broken aluminium for the sliding doors; rooflights 2.2 W/m²K. Insulation and airtightness should run continuously across the junctions with the existing walls, and the new, enlarged openings should be detailed so they do not become cold bridges.	Section 08
7	**The study's false-pitch roof.** Insulate one plane properly, either the flat ceiling or the slopes, with no gap at the wall head. Ventilate any cold roof void above it.	Section 07
8	**Repair the kitchen corner before it is enclosed.** Let the wall dry out, repoint the stone base in lime and make sure the damp-proof course is continuous. • Keep the new terrace paving at least 150 mm below the damp-proof course and the internal floor level, falling away from the house. • Take every downpipe into a drain. • Line the planters so that soil and water are not held against the house walls.	Section 09
9	**The heat pump.** The east-wall position, near where the oil tanks are now, is a sensible one. • The installer should confirm noise at the nearest neighbour (MCS 020), clear space for airflow, and a drain or soakaway for the defrost water. • Size the unit from a room-by-room heat loss calculation on this scheme, after the works. • Plan the cylinder in the cellar, the underfloor heating zones in the new kitchen, living and dining room (and the dining and living rooms if you extend it), and any upsized radiators upstairs.	Section 10

10	**The rooms in the roof.** The main roof is unchanged in the scheme, but the principal bedroom's dormer cheeks were the wettest walls I measured. Consider insulating the dormer cheeks and sloping ceilings from inside while those rooms are disturbed, or at the next re-roofing. At the least, add the loft top-up and an insulated loft hatch to the works.	Sections 05, 07
11	**The bedroom cracks.** Have a structural engineer report on them before the construction drawings are finalised, so that any cause can be designed out. The extension will need an engineer anyway.	Section 09

Working with the architect

The client has confirmed they are happy for me to share this report with the architect. I would be glad to have a call with the architect, with you both on it if you would like, to go through these points as the scheme moves to its Building Regulations stage, and I can review the specification against this report when it is drafted.

12 Improvement options

Every measure I considered for the house, with my view on whether it suits it. Ratings are relative to each other: ●●● high, ●●○ medium, ●○○ low. Costs are indicative 2026 ranges.

MEASURE	SUITABLE?	COMFORT	ENERGY	COST	DISRUPTION	NOTES AND RISKS
Extract ventilation (dMEV in wet rooms), door undercuts	Yes, first	●●●	○○○	£900–£1,600, plus £20–£40 a door	Low	Essential before windows and insulation. The single most important step for the mould.
Rear kitchen corner (downpipe, decking gap, repointing)	Yes, first	●●○	○○○	£300–£900	Low	Stops water getting into the wall. Must come before insulation or a new kitchen.
Gutters and moss	Yes	●○○	○○○	£200–£600 a year	Low	Keep the gutters clear and treat the moss with biocide. Do not jet-wash the tiles.
Loft top-up to 270–300 mm, hatch and ducts	Yes	●●○	●●○	£800–£1,800	Low	Keep the eaves ventilated. Can be done now.
Suspended floor insulation from below, full extent	Yes	●●●	●●●	£6,000–£12,000	Low (from below)	PIR between the joists plus an airtight layer. Keep the void ventilated. Essential under underfloor heating.
Replace all windows and external doors	Yes, with the works	●●●	●●○	£20,000–£35,000 whole house	Medium	With trickle vents. Increases airtightness, so ventilation must be in place.
Positive input ventilation (loft)	Consider	●●○	○○○	£600–£1,200	Low	Helps dry the upstairs. Pair with dMEV in the wet rooms.
MVHR (whole house)	Only if very airtight	●●●	●●○	£8,000–£14,000	High	Ducts to every room. Best fitted only in a deep, whole-house retrofit.
Underfloor heating, ground floor	Yes, with the works	●●●	●○○	£70–£120/m²	Medium	Over the floor insulation. Ideal partner for a heat pump.
Air source heat pump	Yes, after the fabric	●●○	●●●	£14,000–£20,000 before grant	Medium	Sized from a heat loss calculation on the finished spec. New cylinder likely; some radiators may need upsizing.
Heating controls (smart valves, zoning)	Yes	●●○	●○○	£300–£900	Low	Heat rooms when in use, especially the upstairs office.
Rooms-in-roof insulation (slopes, dormer cheeks)	At re-roofing or redecoration	●●●	●●○	£6,000–£15,000	High	Needs careful vapour control. Best from outside when the roof is re-covered.
Cavity wall insulation	Check first	●●○	●●○	£1,500–£3,500	Low	Borescope check first. Exposed site: only a system suited to wind-driven rain.

Internal or external wall insulation	Later, if at all	●●○	●●○	£15,000+	High	Moisture risk in wet rooms; external needs planning permission in the National Landscape.
Solar PV	Possible	○○○	●●○	£6,000–£10,000	Low	South-facing front slope, but the trees will shade it. A shading survey is needed. Useful with a heat pump and EV.

13 Your short- to medium-term plan

This is the order I would do things in, built around the extension and kitchen project rather than separate from it. The principle is simple: first keep the house dry and ventilated, then insulate, then change how it is heated. Each step makes the next one safer and more effective, and the big items are timed to happen while the builders are already on site, so you only pay for disruption and making good once.

Stage 0

NOW · THIS WINTER
BEFORE THE WORKS

MEASURE	COST
Repair the boiler (booked) and heat the house steadily	In hand
Replace the three bathroom fans with dMEV; insulate the ducts in the loft	£900–£1,600
Trim all internal doors to a 10 mm undercut	£20–£40 a door
Rear kitchen corner: downpipe to the drain, decking cut back, repoint	£300–£900
Clean the mould with fungicidal wash; mould-resistant paint in the bathrooms and on the bedroom ceiling	DIY or £300–£700
Clear the gutters; treat the roof moss; check the dormers after heavy rain	£200–£600
Top up the loft to 270–300 mm; insulate and seal the hatch	£800–£1,800
Dehumidifier for laundry and the kitchen; pull furniture off the outside walls	£150–£250
Structural engineer to inspect the bedroom cracks	£350–£600

Stage 1

WINTER 2026–27
DESIGN WITH THE ARCHITECT

DECISION OR DESIGN ITEM	WHO
Add a whole-house ventilation strategy to the architect's scheme: dMEV in the new kitchen, living and dining room, the internal utility, the WC and the bathrooms, with or without PIV; downdraft hob ducted outside	The architect, with me
Floor insulation detail for the whole suspended floor, and the underfloor heating zones (kitchen, dining, and the living room?)	The architect, with me
Window and door schedule for the whole house (all replaced, as drawn), with trickle vents and thermally broken sliding doors	The architect, you
Building Regulations specification for the new rear extension and study: current Part L U-values, the floor over the undercroft, and continuous insulation and airtight junctions to the existing house. Consider insulation behind the new tile hanging on the stair gable	The architect
Engineer's findings on the cracks fed into the design	Engineer, the architect
Heat loss calculation on the post-works specification; heat pump feasibility and quotes	Heat loss specialist or installer
Planning decision (application made with the the architect drawings); quotes for the fabric and building work	The architect, you

Stage 2

2027
THE BUILDING WORKS

MEASURE	COST
Build the architect's scheme: the new kitchen, living and dining extension in place of the conservatory, the replacement study, the utility in the former snug, and the paved terrace and planters. Repair the kitchen corner before it is enclosed	The architect's costing
Insulate the whole suspended floor from below while the trades are on site	£6,000–£12,000
Underfloor heating to the ground floor, over the insulation	£70–£120/m²
Replace all windows and external doors, including the front entrance and its single-glazed top lights	£20,000–£35,000
Kitchen and utility extract to outside; continuous dMEV; PIV if chosen	£800–£2,000
Airtightness detailing at the junctions between new and old, and around every new window	In the build

Stage 3

WITH OR AFTER THE WORKS
HEATING

MEASURE	COST
Air source heat pump sized from the heat loss calculation, with a new cylinder and any radiator upgrades upstairs; remove the oil tanks	£14,000–£20,000 before grant
Smart zoned controls; time-of-use or EV electricity tariff	£300–£900

Stage 4

3–10 YEARS
AT TRIGGER POINTS

MEASURE	TRIGGER
Insulate the roof slopes and dormer cheeks of the rooms in the roof	Re-roofing, or redecorating those rooms
Cavity wall check, and fill if suitable	Any time; borescope first
External wall insulation, if wanted	Re-rendering; planning permission needed
Solar PV, subject to a shading survey	After the heat pump

> Doing things out of order is the most common way retrofit goes wrong. In particular: get the ventilation working before the new windows go in, fix the water at the back corner before insulating or fitting the kitchen against it, and size the heat pump only once the fabric specification is fixed.

Many of these decisions sit with the architect's design; section 11 lists the points to add to his specification.

14 What to avoid, and next steps

Please avoid

- **Blocking air bricks or vents** to stop draughts. The floor void and the rooms need them. Seal the gaps *inside* the rooms, never the vents *outside*.
- **Making the house airtight before the ventilation works**, for example by fitting new windows without trickle vents or before the fans are upgraded.
- **Partial insulation** that stops short and leaves a cold strip, for example insulating under the kitchen but not the rest of the floor.
- **Spray foam on the underside of the roof.** It hides the condition of the timbers, can trap moisture against them, and often causes problems with mortgages and sales.
- **Cement mortar or impermeable coatings** on the old stone plinth; use lime-based repairs there.
- **Sizing a heat pump on today's house.** Wait for the post-works specification.

Next steps

- Read through the report and call me with any questions; a follow-up conversation is included in the fee.
- Get the Stage 0 work done this winter: fans, door undercuts, the back corner, the loft and the engineer.
- Contact one of the structural engineers listed in section 09. I will send you contacts for a heat loss / heat pump specialist.
- Share this report with the architect, and I will arrange a call with him to go through the points in section 11.
- I am happy to review quotes or specifications against this report.

Assessor

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Limitations

This report is advisory. It sets out the condition, moisture and ventilation findings, the measures that appear suitable for the house, the order to carry them out in and the risks to manage. It is not a design, a specification, a structural assessment or an instruction to carry out work. It is not an Energy Performance Certificate or a PAS 2035 retrofit assessment, and nothing has been lodged on any register. The inspection was visual and non-invasive: I did not lift floor coverings or boards, open up walls, ceilings or roof slopes, or test services. The roof was viewed from the ground; the weather did not allow a drone survey, which I have offered to carry out on a return visit. Areas hidden by stored items, boarding, fittings or decoration, including the eaves storage and the void under the 2010 extension, could not be inspected. Humidity and damp readings record conditions on the day, when the heating had been off for about two weeks and the weather was wet; non-invasive damp readings are comparative and indicate where moisture is present rather than measuring it. Costs are indicative 2026 ranges for budgeting only. The report is prepared for the client under the Terms and Conditions of Engagement sent on 19 October 2026, and should not be relied on by anyone else.